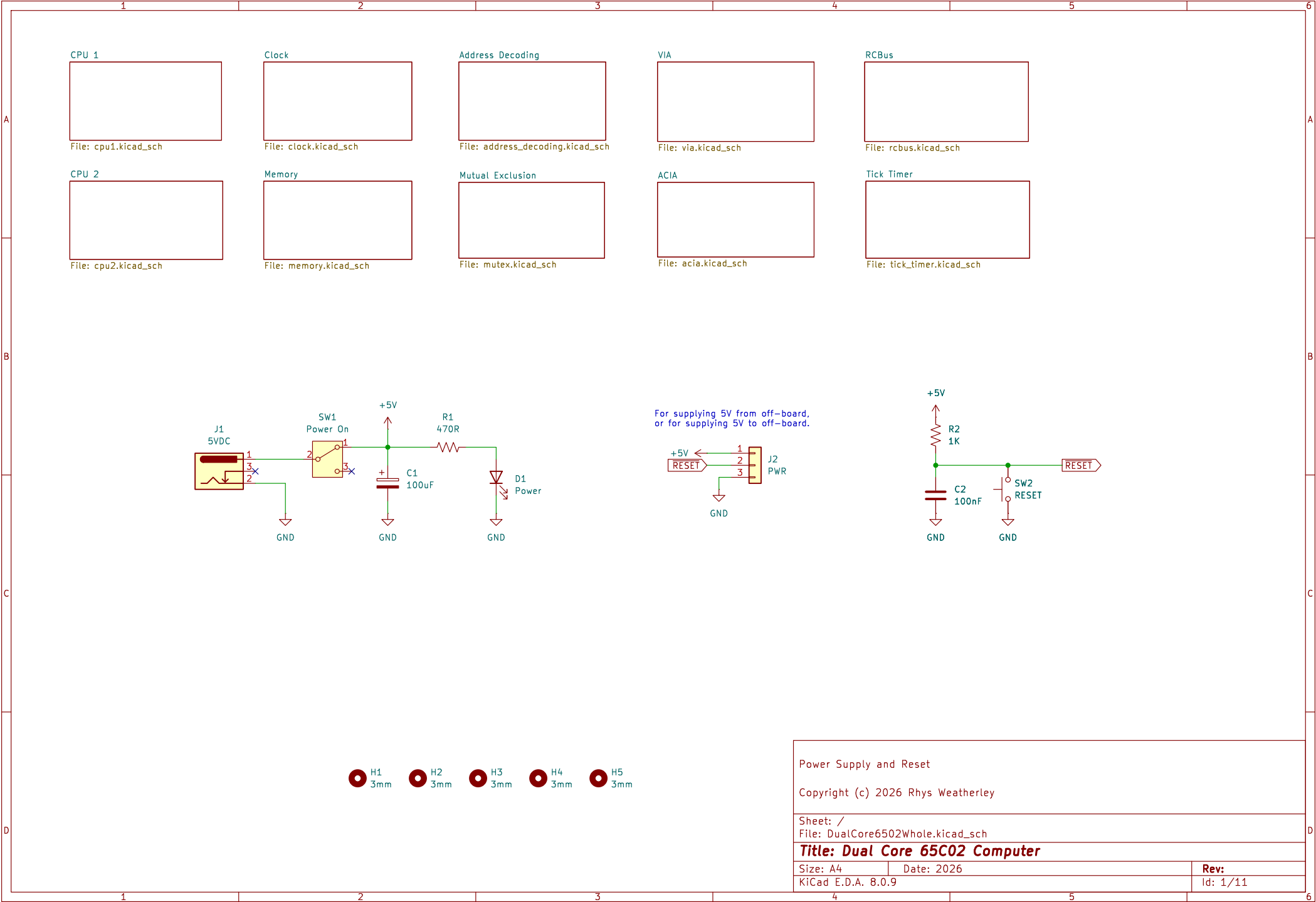
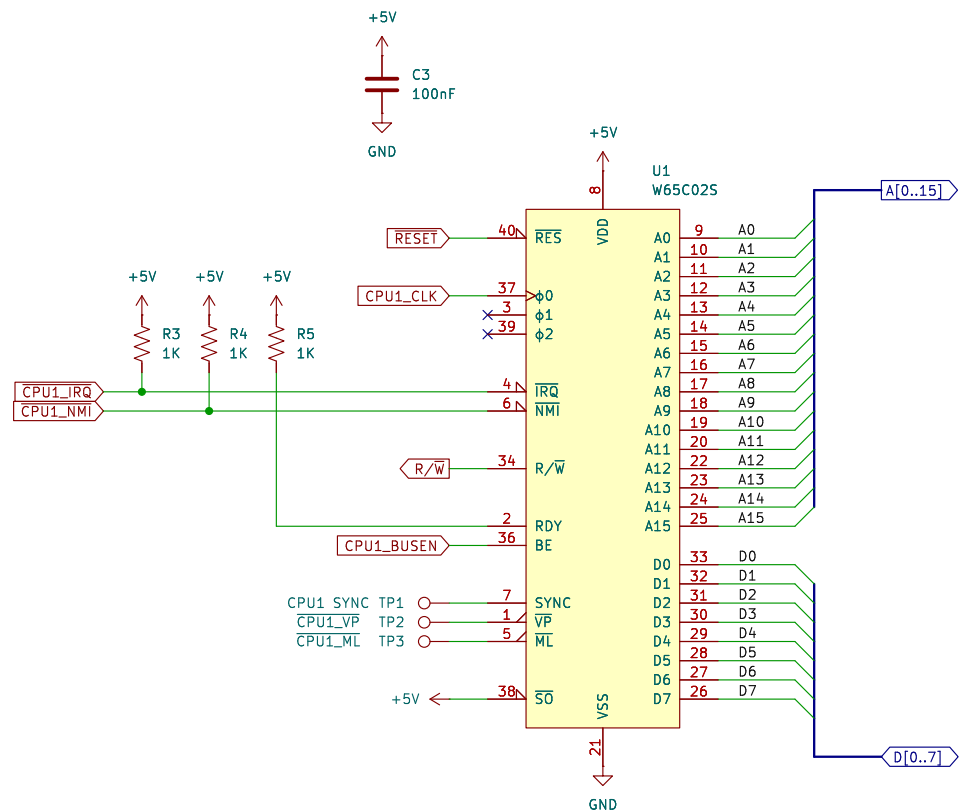






CPU 1

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Sheet: /CPU 1/
File: cpu1.kicad_sch

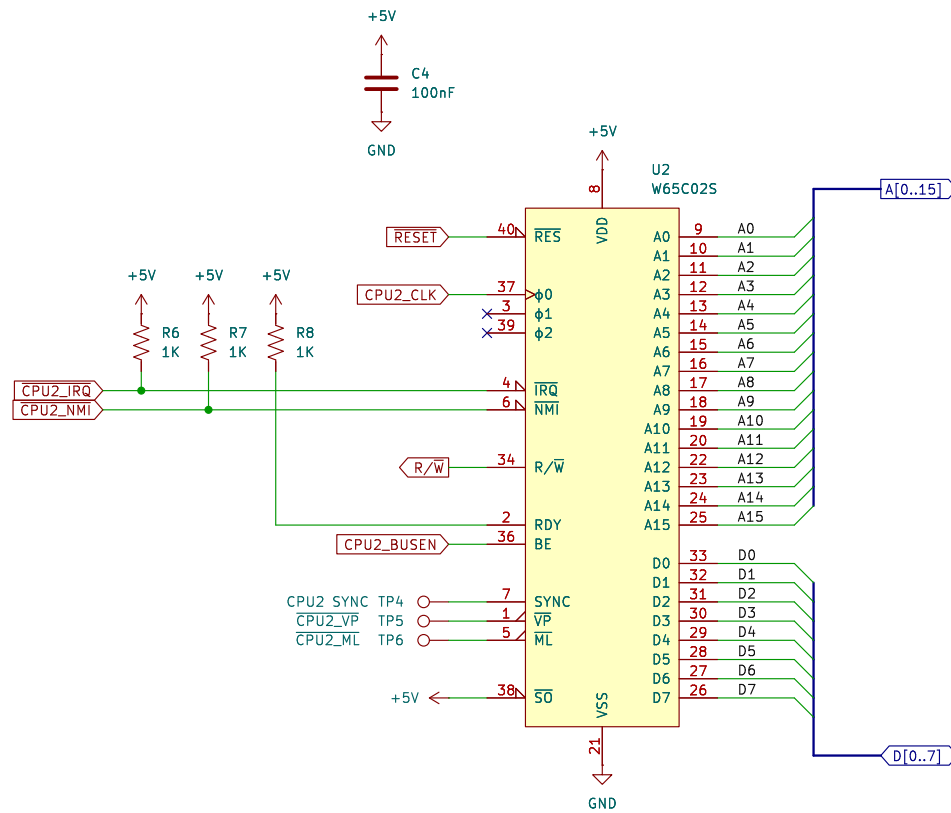
Title: Dual Core 65C02 Computer

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CPU 2

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Sheet: /CPU 2/
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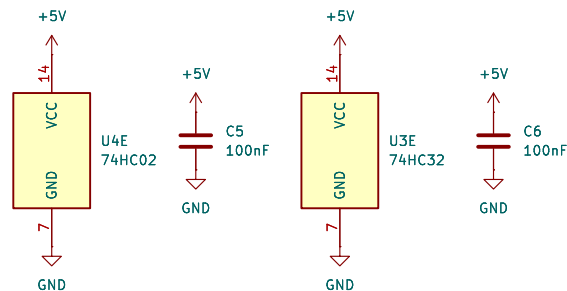
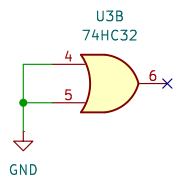
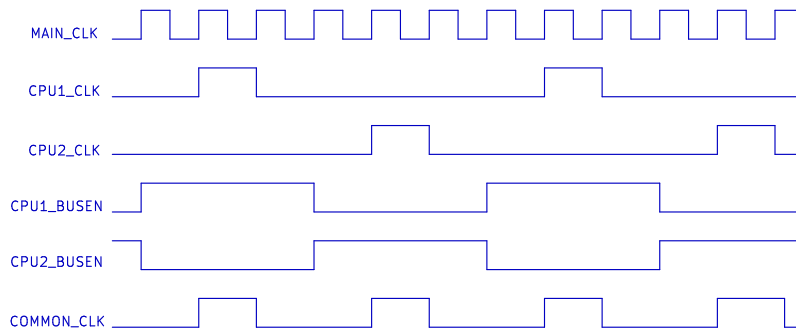
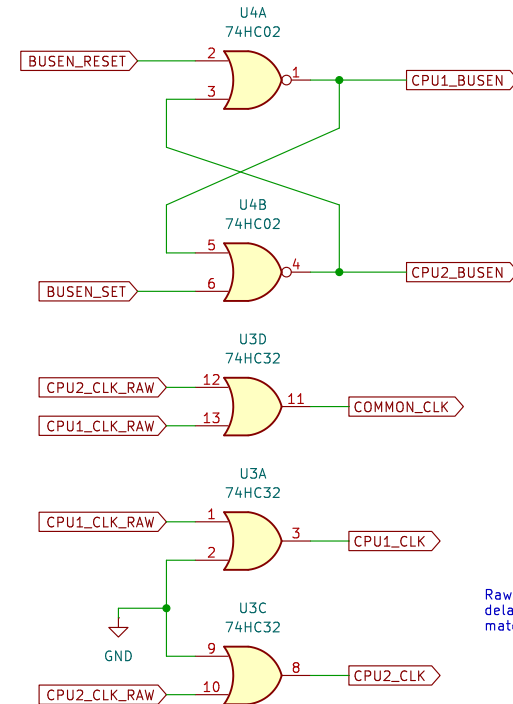
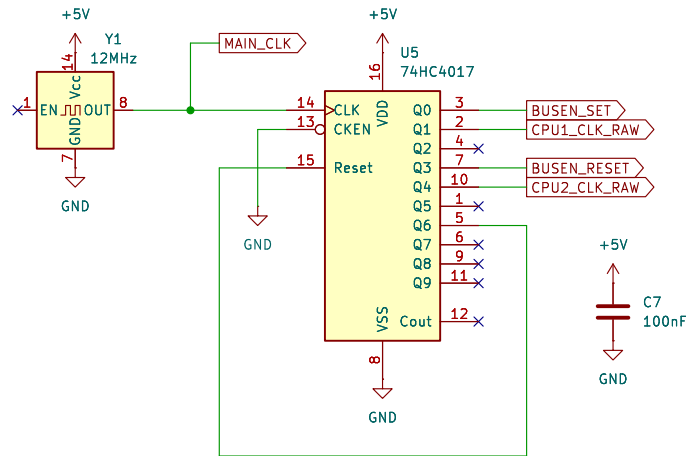
Title: Dual Core 65C02 Computer

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MAIN_CLK (12MHz) is divided by 6 to create CPU1_CLK and CPU2_CLK (2MHz). Also divided by 6 to generate bus enables CPU1_BUSEN and CPU2_BUSEN.

CPU1 can access the common bus only when CPU1_BUSEN is high and CPU2 can access the common bus only when CPU2_BUSEN is high.

Bus enables are longer than clock pulses to allow the CPU's to set up the next bus address prior to the rising edge and to maintain R/W stability.

COMMON_CLK can be used to gate access to daughter chips on either clock.

Clock

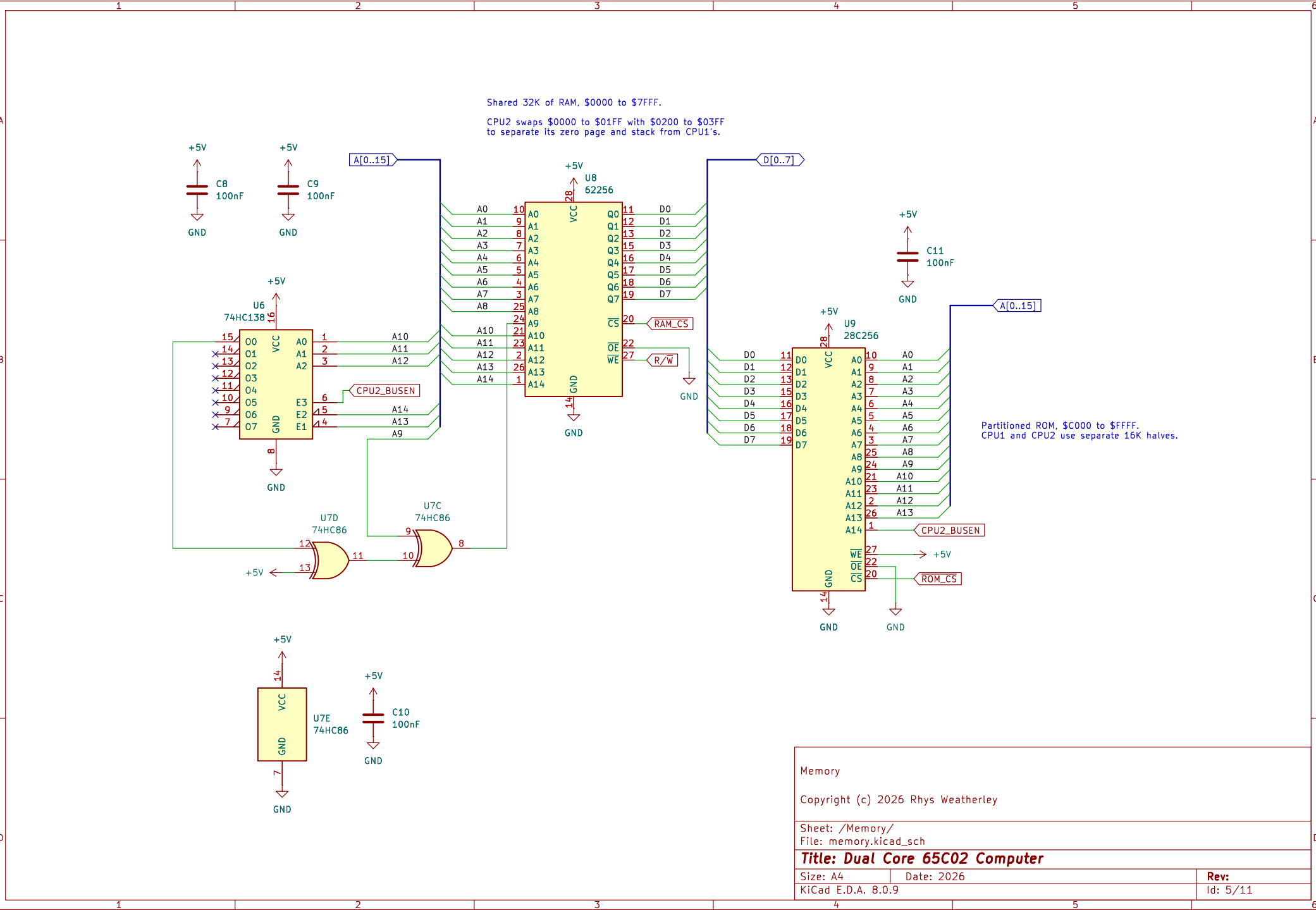
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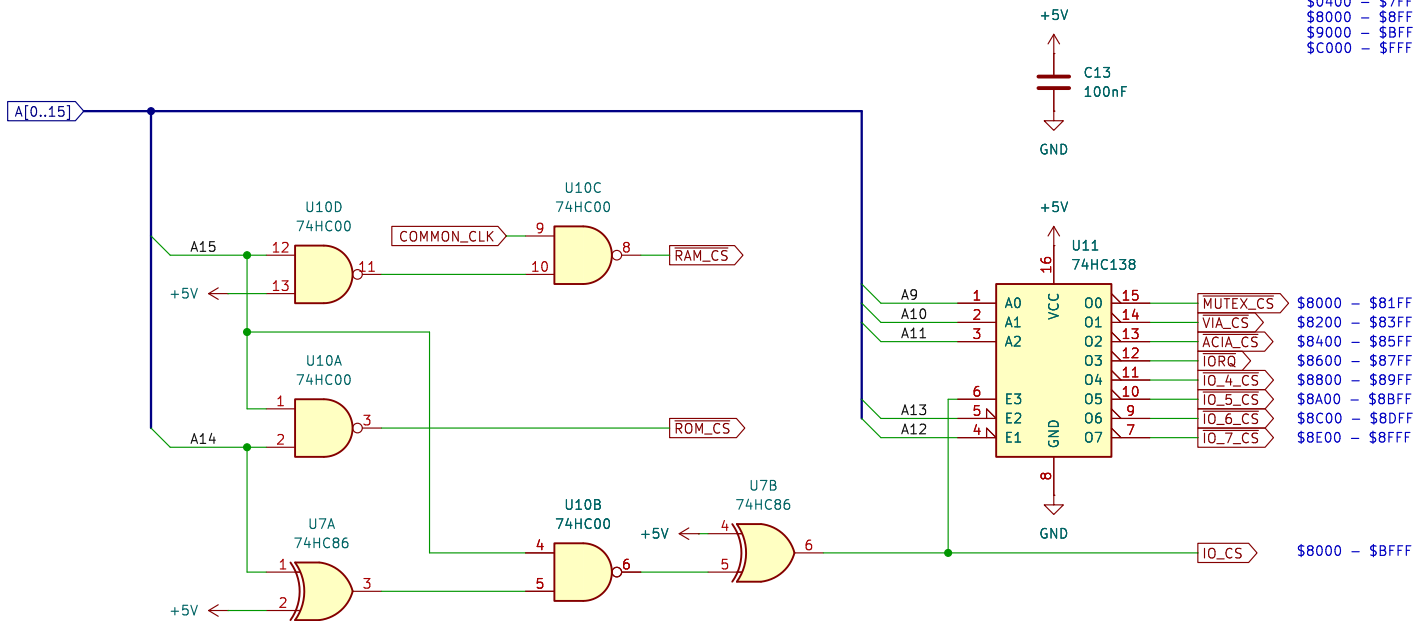
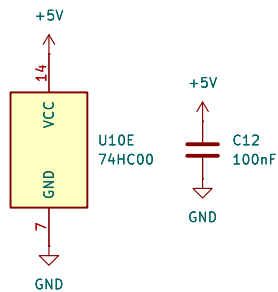
Sheet: /Clock/
File: clock.kicad_sch

Title: Dual Core 65C02 Computer

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Memory Map:

\$0000 - \$01FF - Zero page and stack for CPU1
 \$0200 - \$03FF - Zero page and stack for CPU2
 \$0400 - \$7FFF - Shared RAM
 \$8000 - \$8FFF - I/O Devices
 \$9000 - \$BFFF - Spare
 \$C000 - \$FFFF - This CPU's ROM

Address Decoding

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Sheet: /Address Decoding/
 File: address_decoding.kicad_sch

Title: Dual Core 65C02 Computer

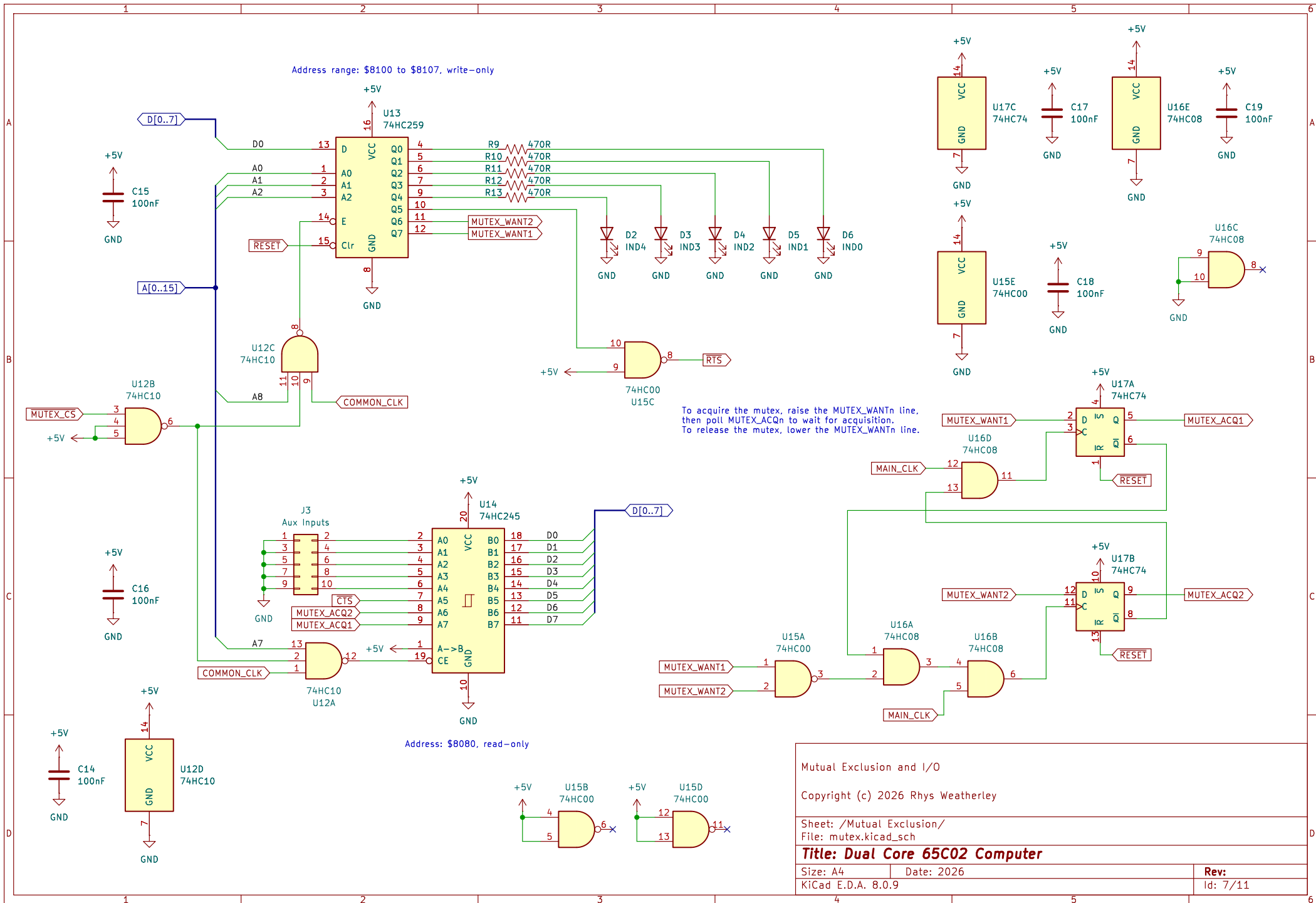
Size: A4

Date: 2026

Rev:

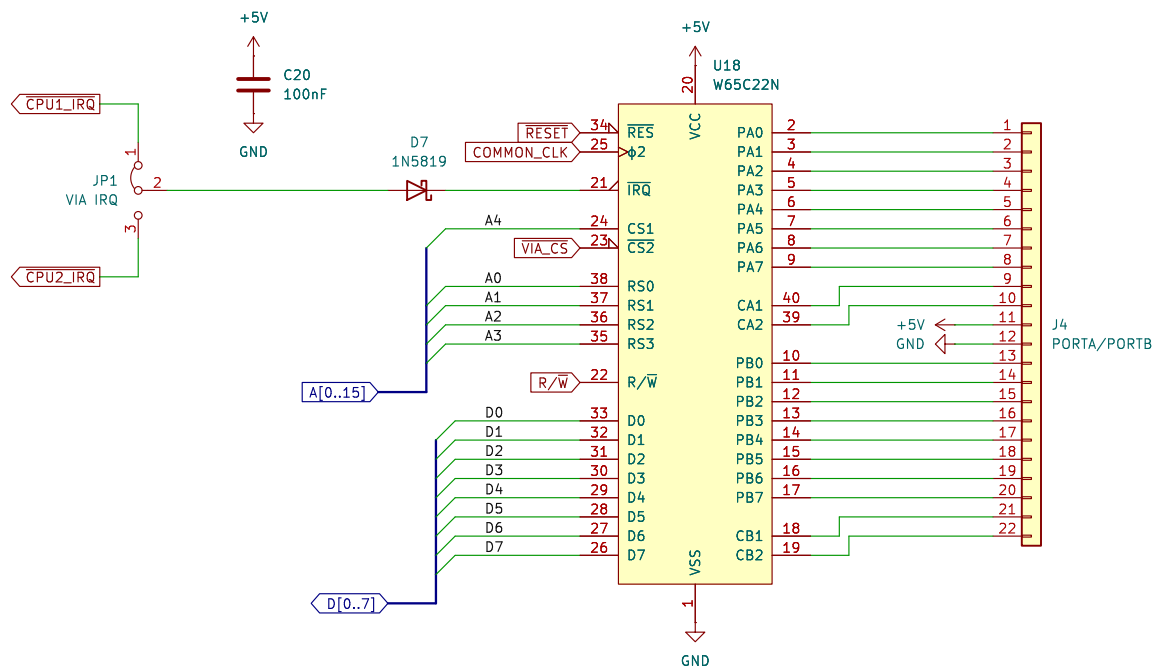
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Address range: \$8210 to \$821F on CPU1

Other VIA's can be placed at \$822x, \$824x, \$828x, and \$830x.
A4 must not be 1 in the other addresses or a bus conflict will occur.



VIA

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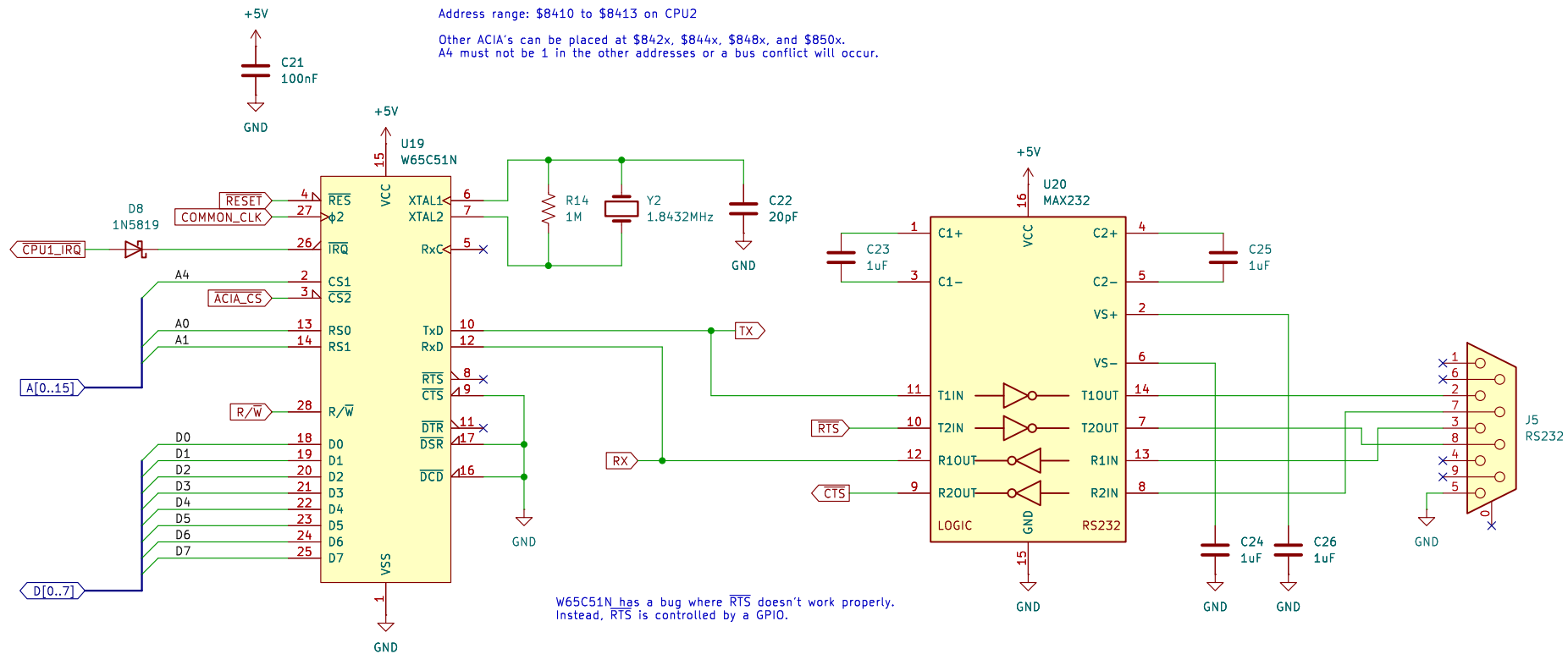
Sheet: /VIA/
File: via.kicad_sch

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ACIA

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Sheet: /ACIA/
 File: acia.kicad_sch

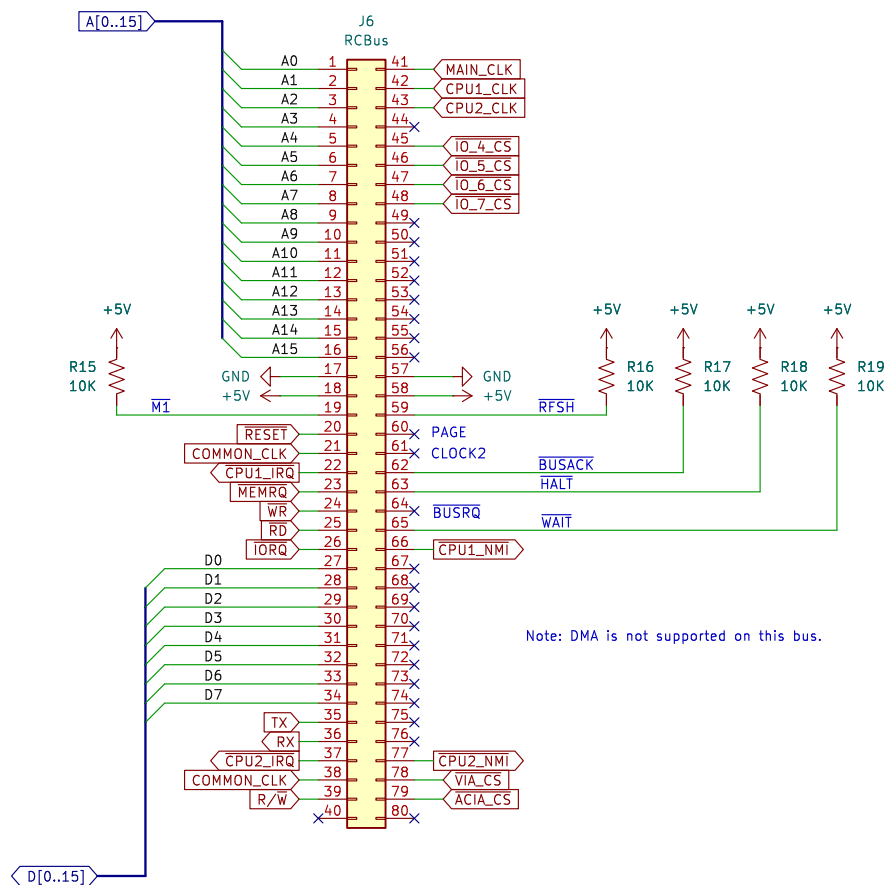
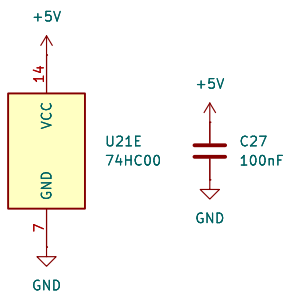
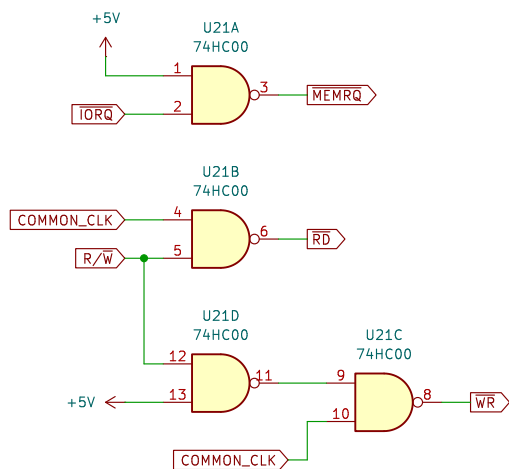
Title: Dual Core 65C02 Computer

Size: A4 Date: 2026

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Rev:
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Address range \$8600 to \$86FF is used for Z80-style I/O.



RCBus Interface

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Invert so CPUn_NMI is pulled high during reset.

1000Hz Tick Timer		
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Title: Dual Core 65C02 Computer		
Size: A4	Date: 2026	Rev:
KiCad E.D.A. 8.0.9		Id: 11/11